Project P1: Steel Production Quality Prediction

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Topic and Motivation

This project focuses on the industrial steel manufacturing process. The main goal is to use machine learning to predict a specific quality factor, called "output," based on various production parameters.

Understanding these relationships is important because it allows companies to maintain high standards and improve the efficiency of their production lines. By comparing over 25 different mathematical models, this study identifies which algorithms are best at handling real-world industrial data.

Related Work

Industrial quality prediction systems can be categorized based on the machine learning methods used to process production data. Current research and applications generally fall into the following groups:

- **Linear and Regularized Models**: Traditional approaches include Linear Regression, Ridge, and Lasso. These methods are valued for their simplicity, but they often struggle with the complex, non-linear nature of industrial datasets, sometimes resulting in poor or negative R^2 scores during testing.
- **Tree-Based and Ensemble Methods**: Advanced systems utilize algorithms like Random Forest, Gradient Boosting, and XGBoost. These models are highly effective at capturing the complicated relationships between 21 different production features and have been shown to outperform linear models in validation performance.
- **Instance-Based and Neural Approaches**: Other systems apply K-Nearest Neighbors (KNN) and Multi-Layer Perceptrons (MLP). While these provide an alternative to tree-based models, their predictive power for quality metrics is often moderate compared to ensemble techniques.

The effectiveness of these systems also depends on data preprocessing strategies. Standard practices involve scaling data and selecting features through techniques like Mutual Information. Furthermore, the use of **Residual Learning** has emerged as a key technique to improve model stability and achieve positive performance results in industrial regression tasks.

In this project, I built and evaluated a complete pipeline comparing 28 different regression models to identify the most accurate method for predicting steel quality using industrial process parameters.

Methods

The methodology implemented in this study follows a comprehensive machine learning pipeline designed to transform industrial production metrics into accurate steel quality predictions. The workflow is divided into five modular stages: data loading, preprocessing, exploratory analysis, model training, and results evaluation.

Data Acquisition and Validation

The dataset utilized for this research originates from an industrial manufacturing environment and consists of 21 numerical features. The primary objective is the prediction of a continuous target variable labeled as `output`, representing a normalized quality metric. During the initial ingestion phase, the data is validated for structural consistency, ensuring that no missing values are present across the around 7000 training samples and 3000 test samples.

Advanced Preprocessing and Feature Selection

To prepare the data for regression, a rigorous preprocessing protocol is followed:

- **Data Partitioning**: The training dataset is split into an 80% training subset and a 20% validation subset. This allows for iterative model tuning without compromising the integrity of the final test set.
- **Feature Prioritization**: Mutual Information (MI) is applied to all 21 features to quantify the statistical dependency between each input variable and the quality target. This non-parametric method is chosen for its ability to capture both linear and non-linear relationships, which are common in complex manufacturing processes.
- **Feature Scaling**: A `StandardScaler` is utilized to normalize the feature ranges. Crucially, the scaler is fitted only on the training data and subsequently applied to the validation and test sets to prevent data leakage.

Residual Learning Framework

A central technical contribution of this methodology is the application of a Residual Learning strategy. During preliminary testing, standard regression models were observed to struggle to produce positive R^2 scores in this specific dataset. To address this, the target variable is transformed into a residual value:

$$y_{\text{residual},i} = y_i - \bar{y}_{\text{train}} \tag{1}$$

where $\bar{y}_{\text{train}}$ represents the mean of the training target. By training models to predict the deviation from the mean rather than the absolute value, the learning process becomes more stable and robust against noise in the production data.

Model Training and Optimization

The project evaluates 28 distinct regression algorithms to ensure a broad exploration of the model space. These models are implemented using libraries such as `scikit-learn` and `XGBoost`. The models are categorized into four groups to compare different mathematical approaches to the problem:

Table 1: Taxonomy of Machine Learning Models Evaluated

Category	**Key Algorithms**	**Focus**
Linear	Ridge, Lasso, ElasticNet	Bias-Variance Trade-off
Tree-Based	Decision Trees	Hierarchical Feature Splitting
Ensemble	Gradient Boosting, Random Forest	Variance and Bias Reduction
Instance/Neural	KNN, MLP	Neighborhood and Layered Mapping

During the training phase, the system tracks not only the predictive accuracy (R^2, RMSE) but also the computational cost, specifically the training and inference times. This multi-metric approach ensures that the final selected model is both accurate and viable for real-time industrial monitoring.

The performance of the steel quality prediction models is assessed through a multi-metric evaluation framework that includes accuracy, error magnitude, and computational efficiency. Each algorithm is rigorously tested using a split-dataset approach, measuring performance on training, validation, and test sets to ensure the robustness of the findings.

Results

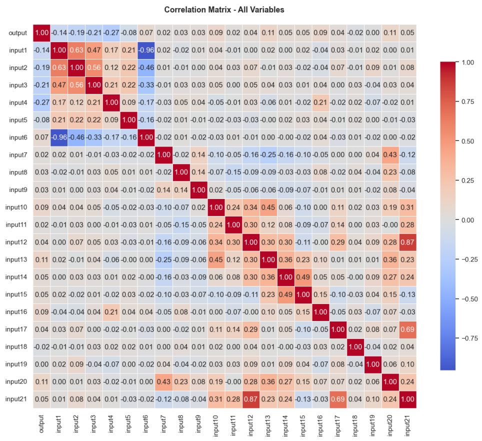
Evaluation Strategy

Model performance was evaluated using the Coefficient of Determination (R^2) and the Root Mean Square Error (RMSE). The R^2 metric quantifies the proportion of variance in steel quality explained by the model, while RMSE measures the average prediction error in the same units as the target variable, allowing for direct interpretability.

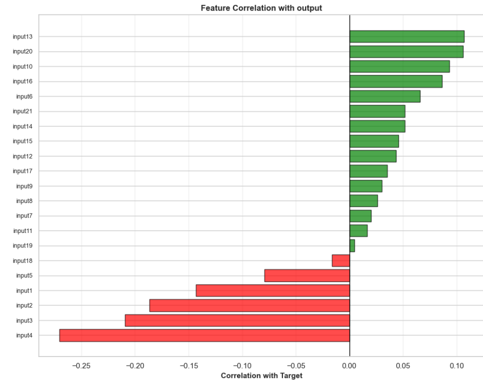
In addition, an *overfitting gap* between training and validation metrics was systematically monitored to ensure that the selected models generalize effectively to unseen industrial production data rather than memorizing the training samples.

Data Characteristics and Exploratory Analysis

Exploratory analysis revealed complex multivariate relationships among the 21 production features. The correlation matrix (Figure 1a) and the feature–target correlation analysis (Figure 1b) show that while several variables exhibit statistical dependence with steel quality, no single feature dominates the prediction process. This observation supports the use of multivariate regression models instead of simple linear approaches.



(a) Correlation matrix of the 21 production features.



(b) Correlation between production features and the steel quality target.

Figure 1: Comparación de matrices de correlación.

The distributions of the features and the target variable were examined using histograms (Figure 2), KDE plots (Figure 2), and the target distribution (Figure 3). The relatively narrow range of the target variable explains the difficulty in achieving high R^2 scores across models.

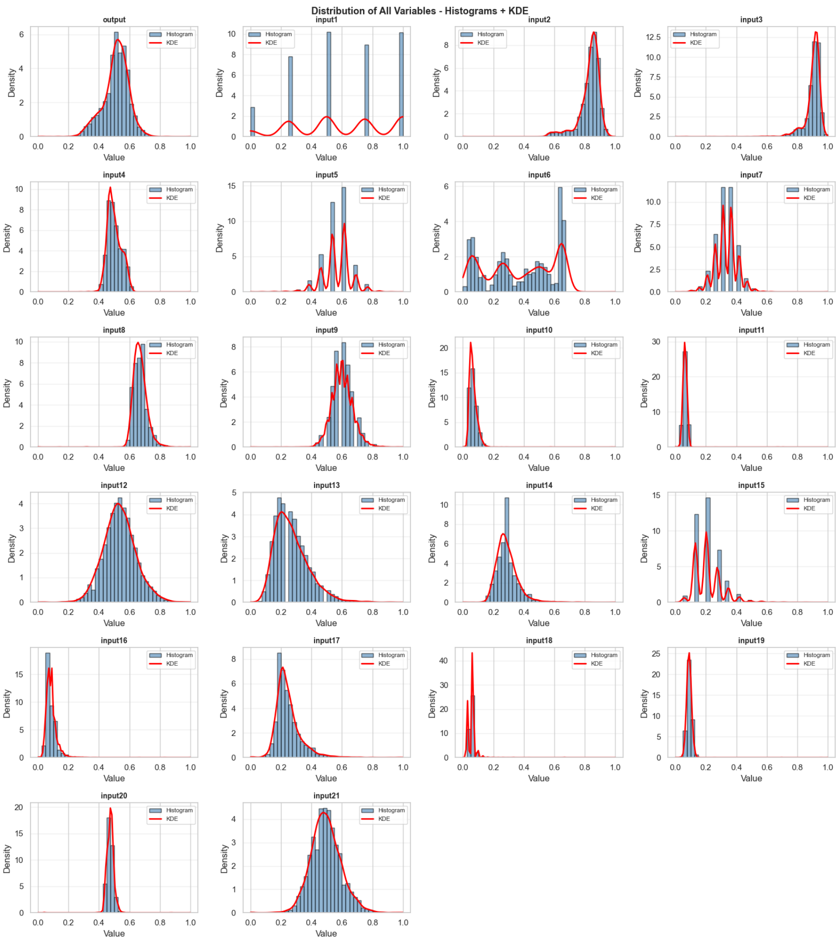


Figure 2: Feature distributions with Kernel Density Estimation (KDE).

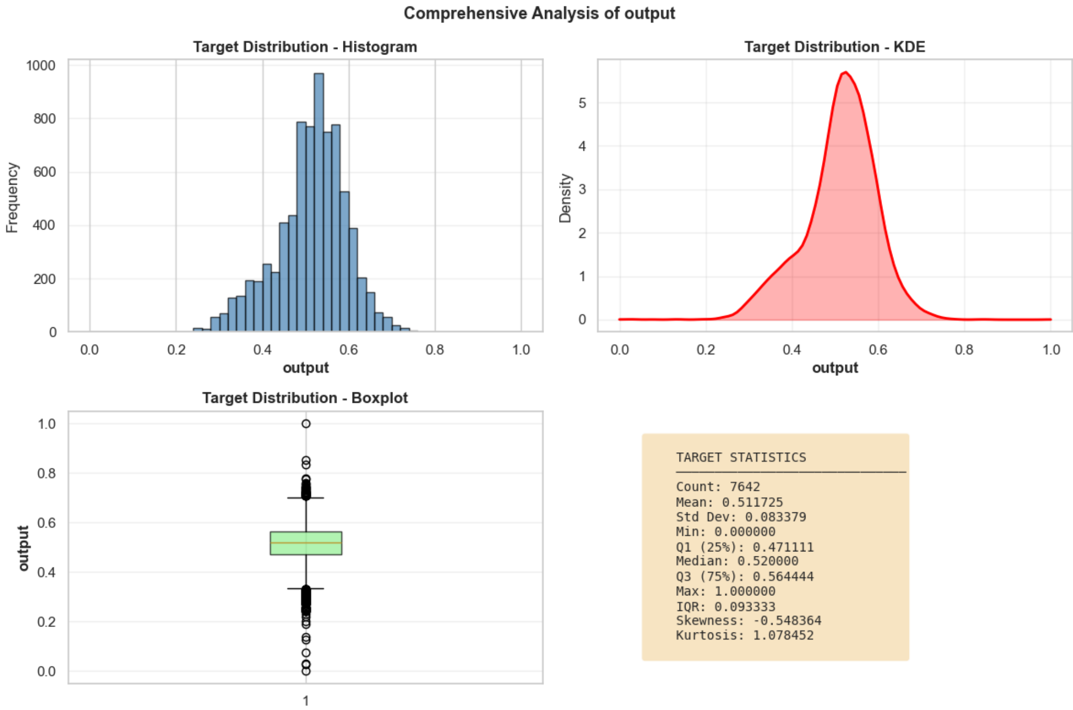


Figure 3: Distribution of the steel quality target variable.

The target variable exhibits a narrow, unimodal, and approximately symmetric distribution concentrated within a limited range, indicating low variance. This characteristic inherently constrains the maximum achievable R^2, as even small absolute prediction errors represent a substantial proportion of the total variance. Furthermore, most input features are bounded within $[0, 1]$ and are heavily concentrated in restricted sub-intervals, reducing effective variability and limiting the explanatory capacity of both linear and nonlinear models. The feature set also displays pronounced heterogeneity in distributional shapes, including approximately Gaussian variables (e.g., `input8`, `input9`, `input12`, `input21`), strongly right-skewed variables with mass near zero (e.g., `input10`, `input11`, `input16--input19`), and multimodal or discrete-like patterns (e.g., `input1`, `input5`, `input6`, `input15`). This diversity suggests the presence of complex and potentially non-smooth relationships between the input variables and the target.

Further analysis of skewness and kurtosis (Figure 4).

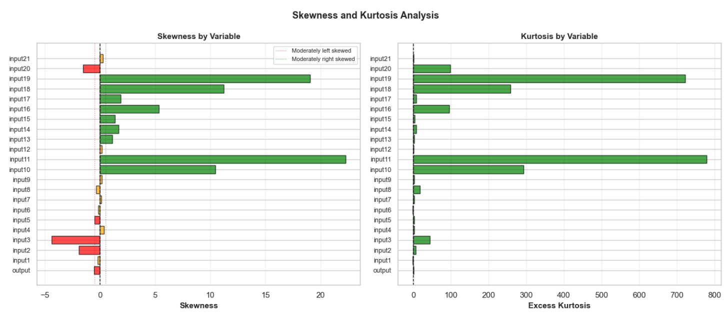
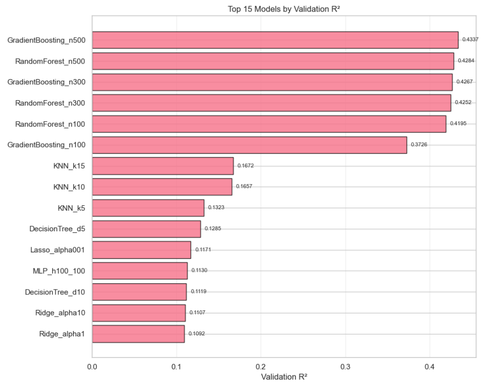


Figure 4: Skewness and kurtosis analysis of production features.

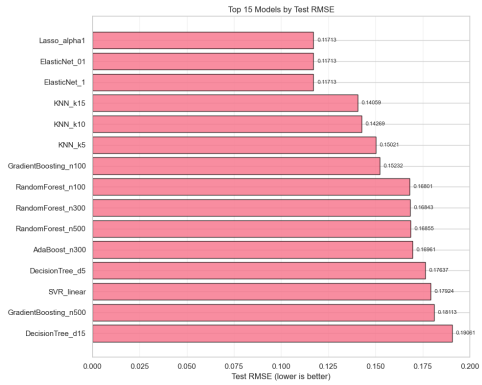
Model Performance Comparison

A total of 28 regression models were trained and evaluated. The results show clear differences between model families: linear models performed poorly across all metrics, while tree-based ensemble methods consistently achieved better predictive results.

When models are ranked by validation R^2 (Figure 5a) and test RMSE (Figure 5b), Gradient Boosting and Random Forest models appear repeatedly among the top performers. This consistency suggests that these methods are better suited to capture the non-linear relationships present in the data.



(a) Top-performing models ranked by validation R^2.



(b) Top-performing models ranked by test RMSE.

Figure 5: Comparison of model performance.

A broader comparison across multiple metrics is presented in Figure 6, which helps illustrate trade-offs between accuracy, generalization, and training time.

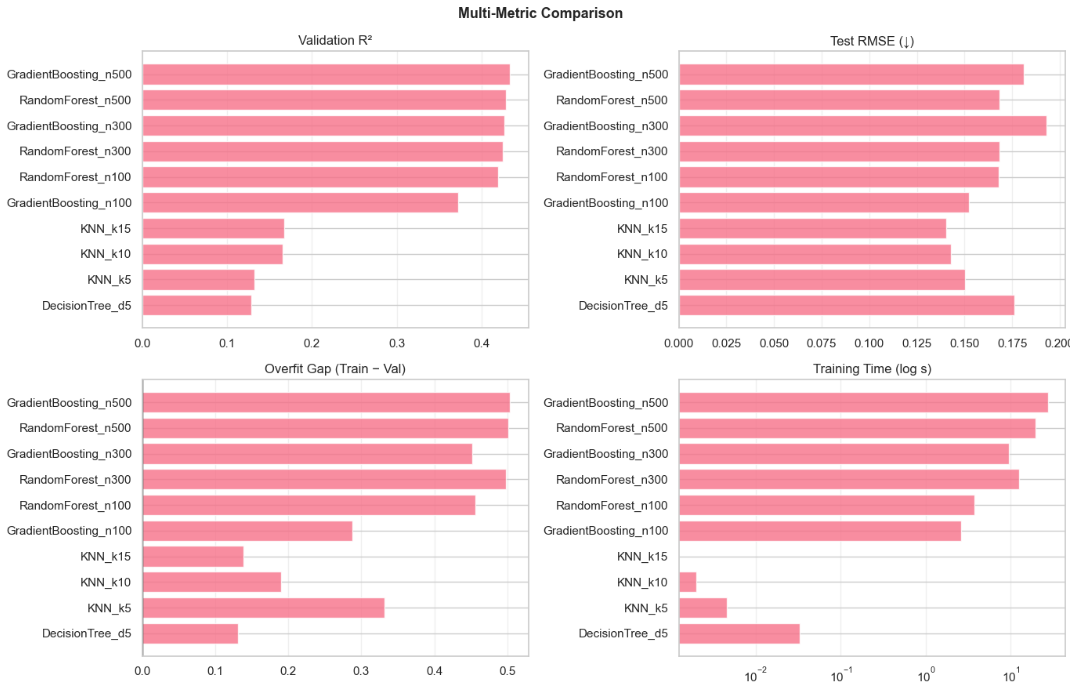


Figure 6: Multi-metric comparison of model performance.

Analysis of Findings

Several practical observations can be drawn from the results:

- **Best overall performance**: Gradient Boosting and Random Forest models achieved the highest validation R^2, with a maximum value close to 0.43.

- **Linear models underperform**: Linear Regression, Ridge, and Lasso showed weak performance and negative test R^2, indicating that they are not flexible enough for this problem.

- **Overfitting behavior**: Many models achieved high training performance but showed a noticeable drop on validation and test data, suggesting overfitting.

- **Training time vs. accuracy**: Gradient Boosting models were the most accurate but also the most computationally expensive, while smaller Random Forest models offered a better balance between accuracy and efficiency.

An important step in achieving stable results was the use of **residual learning**. Without this transformation, several models struggled to learn meaningful patterns between the input features and the target variable.

Detailed Model Assessment

A closer look at the best-performing model is provided in Figure 7. The residuals are mostly centered around zero, indicating limited systematic bias, although the spread of errors remains relatively large.

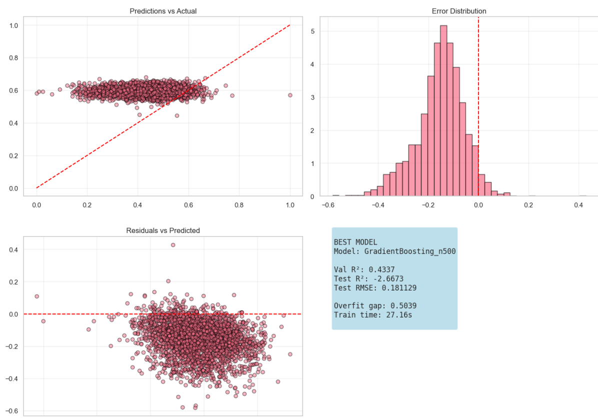


Figure 7: Residual and error analysis of the best-performing model.

Overall, ensemble-based models combined with residual learning provided the most reliable results for steel quality prediction. However, the remaining gap between validation and test performance suggests that further improvements may require better feature engineering and stronger regularization rather than more complex models.